



**EAST TEXAS A&M**  
— UNIVERSITY —

# CSCI 380.61E WEB PROGRAMMING AND INTERFACE

# OVERVIEW

- How to use NodeJS with **Database** (DB)?
- Working with popular database
- **Syntax and packages**
- Examples
- Questions and Discussion

# REVISION QUESTION

Which of the following command will show version of Node?

- A - `$ npm --version`
- B - `$ node --version`
- C - `$ npm getVersion`
- D - `$ node getVersion`

# ANSWER

Answer: **B - \$ node --version**

# REVISION QUESTION

NodeJS is \_\_\_\_\_ Language.

- Server Side
- Client Side
- Both

# ANSWER

Answer: **Server Side**

# REVISION QUESTION

Is Node js multithreaded?

- Yes
- No

# ANSWER

Answer: **No**

# REVISION QUESTION

NodeJS is \_\_\_\_\_

- Asynchronous
- Synchronous

# ANSWER

Answer: **Asynchronous**

# REVISION QUESTION

Which function is used to include modules in Node Js.

- `include();`
- `require();`
- `attach();`

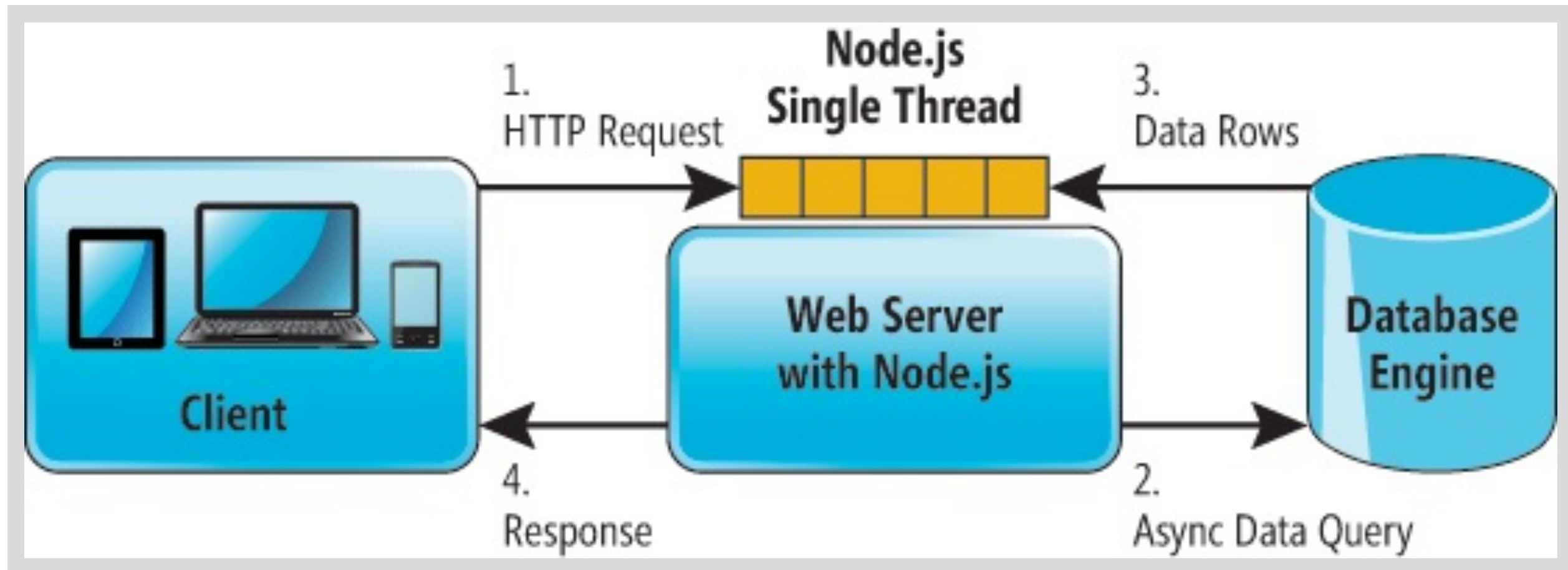
# ANSWER

Answer: **require();**

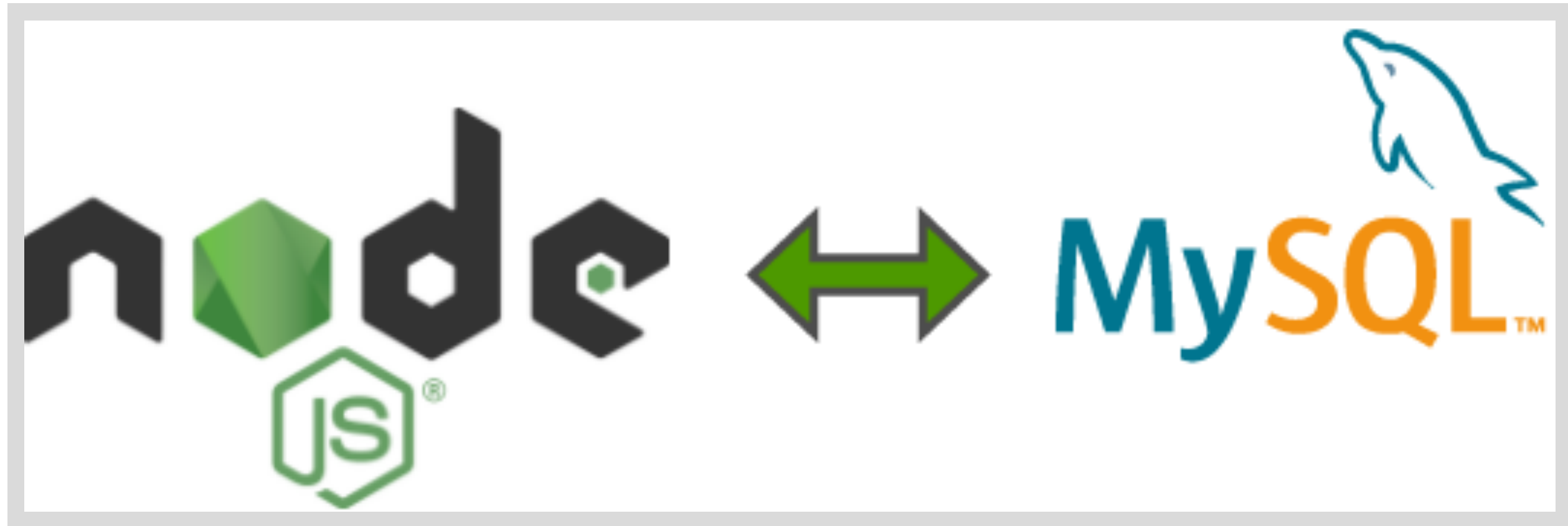
# NODEJS

- NodeJS can be used in **database** applications
- NodeJS supports the majority of popular databases
  - MongoDB, **MySQL**, SQLite, NoSQL, ..

# NODEJS AND DATABASE



# NODEJS WITH MYSQL



# WHY MYSQL?

# WHY MYSQL?

- Highly **reliable**
- Industry **standard**
- On-Demand scalability
- Small footprint
- Cross platform support
- **Free and open-source**

# HOW DO WE USE **MYSQL** WITH **NODEJS?**

# MYSQL WITH NODEJS

- One of the most popular databases is **MySQL**
- Install **"mysql"** module

NodeJS can use this module to manipulate the MySQL database

```
var mysql = require('mysql');
```

# MYSQL WITH NODEJS

- Connect to Database
- Query Database (Check Status)
- Create Database (In Code)
- Update Database
  - Create/Delete/Change Table
  - Insert/Remove/Update Data
- Access Database Data (query)

# CREATE CONNECTION

```
var mysql = require('mysql');

var con = mysql.createConnection({
  host: "localhost",
  user: "yourusername",
  password: "yourpassword"
});

con.connect(function(err) {
  if (err) throw err;
  console.log("Connected!");
});
```

# QUERY DATABASE

```
con.connect(function(err) {  
  if (err) throw err;  
  console.log("Connected!");  
  
  var sql = " ... your SQL query ... "  
  
  con.query(sql, function (err, result) {  
    if (err) throw err;  
    console.log("Result: " + result);  
  });  
});
```

# CREATE DATABASE

```
var mysql = require('mysql');

var con = mysql.createConnection({
  host: "localhost",
  user: "yourusername",
  password: "yourpassword"
});

con.connect(function(err) {
  if (err) throw err;
  console.log("Connected!");
  con.query("CREATE DATABASE mydb", function (err, result) {
    if (err) throw err;
    console.log("Database created");
  });
});
```

# CREATE TABLE

```
var mysql = require('mysql');

var con = mysql.createConnection({
  host: "localhost",
  user: "yourusername",
  password: "yourpassword",
  database: "mydb"
});

con.connect(function(err) {
  if (err) throw err;
  console.log("Connected!");
  var sql = "CREATE TABLE customers (name VARCHAR(255), address VARCHAR(255))";
  con.query(sql, function (err, result) {
    if (err) throw err;
```

# VARIOUS QUERY STATEMENTS

- Create Database
- Create Table
- Insert Into
- Select From
- Where
- Ordered By
- Delete
- Drop Table
- Update
- Limit
- Join

Visit [w3schools details](#)

# SQLFIDDLE - ONLINE SQL SANDBOX

- Easily test out your database commands
- Examples and snippets
- [www.sqlfiddle.com](http://www.sqlfiddle.com)

# WHERE TO START?

- CodeSandbox
  - Setup Basic NodeJS Server/Client
- Setup MySQL Database on FreeSQLDatabase
  - [www.freesqldatabase.com](http://www.freesqldatabase.com)
- Example snippet

```
const mysql = require("mysql");

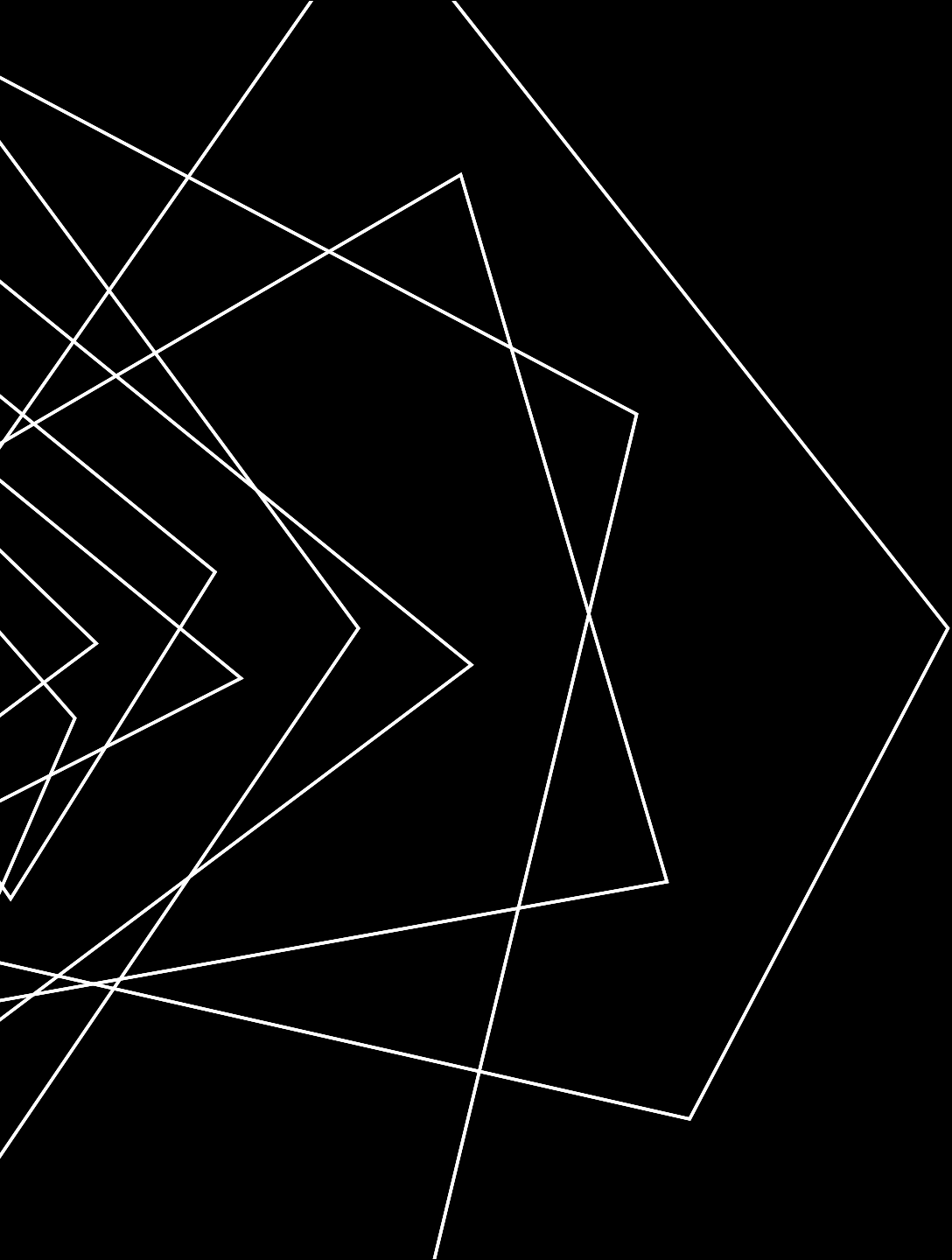
var connection = mysql.createPool({
  connectionLimit: 50,
  host: "sql12.freesqldatabase.com",
  user: "sql12304235",
  password: "1Hj5jAY7Bv",
  database: "sql12304235"
});
```

# THEORY TO PRACTICE

- Important you understand the rational behind NodeJS
- Important you can setup a simple NodeJS server/client
  - e.g., using CodeSandbox
- Important you are able to save/load information from a database
  - MySql
- Important you're actually able to **implement** NodeJS in code (not just the theory)

# SUMMARY

- Understand NodeJS and how to work with a **Database**
- **Manage** a Database (using NodeJS)
- Create, insert and access data using NodeJS and **MySQL**
- **Syntax** and libraries



# THANK YOU

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